

-- VP Planillas Database Schema --

-- === SEQUENCES === --

-- This script creates sequences for various tables in the VP Planillas database.

CREATE SEQUENCE seq_vpg_enterprise_01 START

WITH

1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_branches_02 START

WITH

1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_positions_03 START

WITH

1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_employees_04 START

WITH

1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_payroll_types_05 START

WITH

1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_payrolls_06 START

WITH

1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

```
CREATE SEQUENCE seq_vpg_deductions_07 START  
WITH  
1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
CREATE SEQUENCE seq_vpg_bonuses_08 START  
WITH  
1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
CREATE SEQUENCE seq_vpg_users_09 START  
WITH  
1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
CREATE SEQUENCE seq_vpg_audit_logs_10 START  
WITH  
1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
CREATE SEQUENCE seq_vpg_clock_logs_11 START  
WITH  
1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
CREATE SEQUENCE seq_vpg_labor_events_12 START  
WITH  
1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
CREATE SEQUENCE seq_vpg_report_logs_13 START
```

WITH

1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_report_versions_14 START

WITH

1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_report_targets_15 START

WITH

1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_mail_server_settings_16 START

WITH

1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_employee_documents_17 START

WITH

1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_payrolls_employee_18 START

WITH

1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_employee_labor_event_19 START

WITH

1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

```
CREATE SEQUENCE seq_vpg_vacations_20 START  
WITH  
1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
-- === END OF SEQUENCES === --
```

```
-- === TABLES === --
```

```
-- === Enterprise Structure === --
```

```
CREATE TABLE
```

```
VPG_ENTERPRISE (
```

```
enterprise_id integer primary key default nextval('seq_vpg_enterprise_01'),
```

```
enterprise_name varchar(50) not null,
```

```
enterprise_image bytea not null,
```

```
enterprise_creation_date date not null,
```

```
enterprise_version integer default 1 not null
```

```
);
```

```
CREATE TABLE
```

```
VPG_BRANCHES (
```

```
branch_id integer primary key default nextval('seq_vpg_branches_02'),
```

```
branch_name varchar(50) not null,
```

```
branch_location varchar(100) not null,
```

```
branch_version integer default 1 not null
```

```
);
```

```
CREATE TABLE
```

```
VPG_ENTERPRISE_BRANCH (branch_enterprise_enterprise_id INTEGER,  
branch_enterprise_branch_id INTEGER);
```

```
CREATE TABLE
```

```
VPG_BRANCH_EMPLOYEE (employee_branch_branch_id INTEGER,  
employee_branch_employee_id INTEGER);
```

```
CREATE TABLE
```

```
vpg_positions (  
position_id integer primary key default nextval ('seq_vpg_positions_03'),  
position_name varchar(50) not null,  
position_description text not null,  
position_base_salary decimal(10, 2) not null,  
position_version integer default 1 not null  
);
```

```
-- === Employee Management === --
```

```
CREATE TABLE
```

```
vpg_employees (  
employee_id integer primary key default nextval ('seq_vpg_employees_04'),  
employee_first_name varchar(50) not null,  
employee_last_name varchar(50) not null,  
employee_middle_name varchar(50) not null,  
employee_national_id varchar(30) not null,  
employee_social_code varchar(100) not null,  
employee_email varchar(100) not null,
```

```
employee_position_id integer not null,  
employee_hire_date date not null,  
employee_exit_date date,  
employee_fired boolean default false not null,  
employee_status char(1) not null check (employee_status in ('A', 'I', 'T')),  
employee_version integer default 1 not null  
);
```

```
CREATE TABLE
```

```
vpg_employee_documents (  
employee_documents_id integer primary key default nextval  
('seq_vpg_employee_documents_17'),  
employee_documents_employee_id integer not null,  
employee_documents_file_path varchar(255) not null,  
employee_documents_document_type varchar(50) not null,  
employee_documents_uploaded_at timestamp not null  
);
```

```
-- === Payroll and Deductions Management === --
```

```
CREATE TABLE
```

```
vpg_payroll_types (  
payroll_types_id integer primary key default nextval ('seq_vpg_payroll_types_05'),  
payroll_types_name varchar(50) not null,  
payroll_types_description text not null,  
payroll_types_version integer default 1 not null  
);
```

CREATE TABLE

vpg_payrolls (

payrolls_id integer primary key default nextval ('seq_vpg_payrolls_06'),

payrolls_payroll_type_id integer not null,

payrolls_period_start date not null,

payrolls_period_end date not null,

payrolls_payment_date date not null,

payrolls_status varchar(20) not null,

payrolls_version integer default 1 not null

);

CREATE TABLE

vpg_payroll_employee (

payroll_employee_id integer primary key default nextval
('seq_vpg_payrolls_employee_18'),

payroll_employee_payroll_id integer not null,

payroll_employee_employee_id integer not null,

payroll_employee_gross_salary decimal(10, 2) not null,

payroll_employee_total_deductions decimal(10, 2) not null,

payroll_employee_net_salary decimal(10, 2) not null,

payroll_employee_version integer default 1 not null

);

CREATE TABLE

vpg_deductions (

```
deductions_id integer primary key default nextval ('seq_vpg_deductions_07'),
deductions_name varchar(100) not null,
deductions_description text not null,
deductions_percentage decimal(5, 2),
deductions_fixed_amount decimal(10, 2),
deductions_version integer default 1 not null
);
```

CREATE TABLE

```
vpg_vacations (
vacations_id INTEGER PRIMARY KEY DEFAULT nextval ('seq_vpg_vacations_20'),
vacations_employee_id INTEGER NOT NULL,
vacations_start_date DATE NOT NULL,
vacations_end_date DATE NOT NULL,
vacations_total_days INTEGER GENERATED ALWAYS AS (vacations_end_date -
vacations_start_date + 1) STORED,
vacations_paid BOOLEAN DEFAULT TRUE,
vacations_status VARCHAR(20) DEFAULT 'Aprobado', -- Pendiente, Aprobado, Rechazado
vacations_version INTEGER DEFAULT 1 NOT NULL
);
```

CREATE TABLE

```
vpg_employee_deductions (
employee_deductions_employee_id INTEGER NOT NULL,
employee_deductions_deduction_id INTEGER NOT NULL,
```



```

employee_deductions_payroll_id INTEGER NOT NULL, -- A cuál planilla pertenece esta deducción

employee_deductions_year INTEGER NOT NULL, -- Año fiscal

employee_deductions_month INTEGER NOT NULL, -- Mes de la deducción

employee_deductions_amount DECIMAL(10, 2) NOT NULL, -- Monto aplicado al empleado en esta planilla

employee_deductions_version INTEGER DEFAULT 1 NOT NULL,

PRIMARY KEY (employee_deductions_employee_id, employee_deductions_deduction_id, employee_deductions_payroll_id)

);

```

```

CREATE TABLE

```

```

vpg_bonuses (

bonuses_id INTEGER PRIMARY KEY DEFAULT nextval ('seq_vpg_bonuses_08'),

bonuses_employee_id INTEGER NOT NULL,

bonuses_payroll_id INTEGER NOT NULL, -- Planilla en la que se pagó el bono

bonuses_year INTEGER NOT NULL, -- Año contable

bonuses_month INTEGER NOT NULL, -- Mes contable

bonuses_description VARCHAR(255) NOT NULL,

bonuses_amount DECIMAL(10, 2) NOT NULL,

bonuses_granted_at DATE NOT NULL, -- Fecha en que se concedió

bonuses_version INTEGER DEFAULT 1 NOT NULL

);

```

```

-- === Users and Access Control === --

```

```

CREATE TABLE

```

```

vpg_users (

```

```
user_id integer primary key default nextval ('seq_vpg_users_09'),
user_first_name varchar(50) not null,
user_last_name varchar(50) not null,
user_middle_name varchar(50) not null,
user_national_id varchar(30) not null,
user_email varchar(100) not null,
user_username varchar(50) not null,
user_password varchar(255) not null,
user_role varchar(20) not null,
user_version integer default 1 not null
);
```

CREATE TABLE

```
vpg_audit_logs (
audit_logs_id integer primary key default nextval ('seq_vpg_audit_logs_10'),
audit_logs_user_id integer not null,
audit_logs_action varchar(100) not null,
audit_logs_entity varchar(100) not null,
audit_logs_entity_id integer not null,
audit_logs_timestamp timestamp not null,
audit_logs_details text
);
```

-- === Clock/Attendance Management === --

CREATE TABLE

```
vpg_clock_logs (
```

```
clock_logs_id integer primary key default nextval ('seq_vpg_clock_logs_11'),
clock_logs_employee_id integer not null,
clock_logs_timestamp timestamp not null,
clock_logs_log_type varchar(10) not null,
clock_logs_remarks text,
clock_logs_version integer default 1 not null
);
```

```
-- === Labor Events Management === --
```

```
CREATE TABLE
```

```
vpg_labor_events (
labor_events_id integer primary key default nextval ('seq_vpg_labor_events_12'),
labor_events_name varchar(100) not null,
labor_events_description text not null,
labor_events_version integer default 1 not null
);
```

```
CREATE TABLE
```

```
vpg_employee_labor_event (
employee_labor_event_id integer primary key default nextval
('seq_vpg_employee_labor_event_19'),
employee_labor_event_employee_id integer not null,
employee_labor_event_labor_event_id integer not null,
employee_labor_event_start_date date not null,
employee_labor_event_end_date date,
employee_labor_event_status varchar(20) not null,
```

```
employee_labor_event_version integer default 1 not null  
);
```

```
-- === Report Management === --
```

```
CREATE TABLE
```

```
vpg_report_logs (  
report_logs_id integer primary key default nextval ('seq_vpg_report_logs_13'),  
report_logs_report_type varchar(50) not null,  
report_logs_generated_by integer not null,  
report_logs_generated_at timestamp not null,  
report_logs_period_start date not null,  
report_logs_period_end date not null,  
report_logs_file_path varchar(255) not null,  
report_logs_status varchar(20) not null,  
report_logs_version integer default 1 not null  
);
```

```
CREATE TABLE
```

```
vpg_report_versions (  
report_versions_id integer primary key default nextval ('seq_vpg_report_versions_14'),  
report_versions_report_log_id integer not null,  
report_versions_created_at timestamp not null,  
report_versions_file_path varchar(255) not null,  
report_versions_remarks text  
);
```

CREATE TABLE

```
vpg_report_targets (  
  report_targets_id integer primary key default nextval ('seq_vpg_report_targets_15'),  
  report_targets_institution varchar(100) not null,  
  report_targets_endpoint_url varchar(255) not null,  
  report_targets_auth_token varchar(255) not null,  
  report_targets_contact_email varchar(100) not null,  
  report_targets_version integer default 1 not null  
);
```

-- === Mail Server Settings === --

CREATE TABLE

```
vpg_mail_server_settings (  
  mail_server_settings_id integer primary key default nextval  
  ('seq_vpg_mail_server_settings_16'),  
  mail_server_settings_host varchar(100) not null,  
  mail_server_settings_port integer not null,  
  mail_server_settings_username varchar(100) not null,  
  mail_server_settings_password varchar(255) not null,  
  mail_server_settings_from_address varchar(100) not null,  
  mail_server_settings_use_ssl boolean not null,  
  mail_server_settings_use_tls boolean not null,  
  mail_server_settings_version integer default 1 not null  
);
```

-- === END OF TABLES === --

-- === Relationships === --

-- Enterprise and Branch relationships

```
alter table VPG_ENTERPRISE_BRANCH add constraint  
fk_vpg_enterprise_branch_enterprise_01 foreign key (branch_enterprise_enterprise_id)  
references VPG_ENTERPRISE (enterprise_id);
```

```
alter table VPG_ENTERPRISE_BRANCH add constraint  
fk_vpg_enterprise_branch_branches_02 foreign key (branch_enterprise_branch_id)  
references VPG_BRANCHES (branch_id);
```

```
alter table VPG_BRANCH_EMPLOYEE add constraint  
fk_vpg_branch_employee_branches_03 foreign key (employee_branch_branch_id)  
references VPG_BRANCHES (branch_id);
```

```
alter table VPG_BRANCH_EMPLOYEE add constraint  
fk_vpg_branch_employee_employees_04 foreign key (employee_branch_employee_id)  
references vpg_employees (employee_id);
```

-- Employee relationships

```
alter table vpg_employees add constraint fk_vpg_employees_positions_05 foreign key  
(employee_position_id) references vpg_positions (position_id);
```

```
alter table vpg_employee_documents add constraint  
fk_vpg_employee_documents_employees_06 foreign key  
(employee_documents_employee_id) references vpg_employees (employee_id);
```

-- Payroll relationships

```
alter table vpg_payrolls add constraint fk_vpg_payrolls_payroll_types_07 foreign key  
(payrolls_payroll_type_id) references vpg_payroll_types (payroll_types_id);
```

```
alter table vpg_payroll_employee add constraint fk_vpg_payroll_employee_payrolls_08  
foreign key (payroll_employee_payroll_id) references vpg_payrolls (payrolls_id);
```

```
alter table vpg_payroll_employee add constraint fk_vpg_payroll_employee_employees_09  
foreign key (payroll_employee_employee_id) references vpg_employees (employee_id);
```

-- Deductions relationships

```
alter table vpg_employee_deductions add constraint  
fk_vpg_employee_deductions_employees_10 foreign key  
(employee_deductions_employee_id) references vpg_employees (employee_id);
```

```
alter table vpg_employee_deductions add constraint  
fk_vpg_employee_deductions_deductions_11 foreign key  
(employee_deductions_deduction_id) references vpg_deductions (deductions_id);
```

```
alter table vpg_employee_deductions add constraint  
fk_vpg_employee_deductions_payrolls_12 foreign key (employee_deductions_payroll_id)  
references vpg_payrolls (payrolls_id);
```

-- Bonuses relationships

```
alter table vpg_bonuses add constraint fk_vpg_bonuses_employees_13 foreign key  
(bonuses_employee_id) references vpg_employees (employee_id);
```

```
alter table vpg_bonuses add constraint fk_vpg_bonuses_payrolls_14 foreign key  
(bonuses_payroll_id) references vpg_payrolls (payrolls_id);
```

-- Vacations relationships

```
alter table vpg_vacations add constraint fk_vpg_vacations_employees_15 foreign key  
(vacations_employee_id) references vpg_employees (employee_id);
```

-- Clock logs relationships

```
alter table vpg_clock_logs add constraint fk_vpg_clock_logs_employees_16 foreign key
(clock_logs_employee_id) references vpg_employees (employee_id);
```

-- Labor events relationships

```
alter table vpg_employee_labor_event add constraint
fk_vpg_employee_labor_event_employees_17 foreign key
(employee_labor_event_employee_id) references vpg_employees (employee_id);
```

```
alter table vpg_employee_labor_event add constraint
fk_vpg_employee_labor_event_labor_events_18 foreign key
(employee_labor_event_labor_event_id) references vpg_labor_events (labor_events_id);
```

-- Report relationships

```
alter table vpg_report_logs add constraint fk_vpg_report_logs_users_19 foreign key
(report_logs_generated_by) references vpg_users (user_id);
```

```
alter table vpg_report_versions add constraint fk_vpg_report_versions_report_logs_20
foreign key (report_versions_report_log_id) references vpg_report_logs (report_logs_id);
```

-- Audit logs relationships

```
alter table vpg_audit_logs add constraint fk_vpg_audit_logs_users_21 foreign key
(audit_logs_user_id) references vpg_users (user_id);
```

-- === END OF RELATIONSHIPS === --

-- === Indexes === --

-- Enterprise and Branch indexes


```
create index idx_vpg_enterprise_branch_enterprise_id on VPG_ENTERPRISE_BRANCH  
(branch_enterprise_enterprise_id);
```

```
create index idx_vpg_enterprise_branch_branch_id on VPG_ENTERPRISE_BRANCH  
(branch_enterprise_branch_id);
```

```
create index idx_vpg_branch_employee_branch_id on VPG_BRANCH_EMPLOYEE  
(employee_branch_branch_id);
```

```
create index idx_vpg_branch_employee_employee_id on VPG_BRANCH_EMPLOYEE  
(employee_branch_employee_id);
```

-- Employee indexes

```
create index idx_vpg_employees_position_id on vpg_employees (employee_position_id);
```

```
create index idx_vpg_employees_national_id on vpg_employees (employee_national_id);
```

```
create index idx_vpg_employees_email on vpg_employees (employee_email);
```

```
create index idx_vpg_employees_status on vpg_employees (employee_status);
```

```
create index idx_vpg_employee_documents_employee_id on vpg_employee_documents  
(employee_documents_employee_id);
```

```
create index idx_vpg_employee_documents_type on vpg_employee_documents  
(employee_documents_document_type);
```

-- Payroll indexes

```
create index idx_vpg_payrolls_payroll_type_id on vpg_payrolls (payrolls_payroll_type_id);
```

```
create index idx_vpg_payrolls_period_start on vpg_payrolls (payrolls_period_start);
```

```
create index idx_vpg_payrolls_period_end on vpg_payrolls (payrolls_period_end);
```

```
create index idx_vpg_payrolls_status on vpg_payrolls (payrolls_status);
```

```
create index idx_vpg_payroll_employee_payroll_id on vpg_payroll_employee  
(payroll_employee_payroll_id);
```

```
create index idx_vpg_payroll_employee_employee_id on vpg_payroll_employee  
(payroll_employee_employee_id);
```

```
create index idx_vpg_payroll_employee_composite on vpg_payroll_employee  
(payroll_employee_payroll_id, payroll_employee_employee_id);
```

-- Deductions indexes

```
create index idx_vpg_employee_deductions_employee_id on vpg_employee_deductions  
(employee_deductions_employee_id);
```

```
create index idx_vpg_employee_deductions_deduction_id on vpg_employee_deductions  
(employee_deductions_deduction_id);
```

```
create index idx_vpg_employee_deductions_payroll_id on vpg_employee_deductions  
(employee_deductions_payroll_id);
```

```
create index idx_vpg_employee_deductions_year_month on vpg_employee_deductions  
(employee_deductions_year, employee_deductions_month);
```

-- Bonuses indexes

```
create index idx_vpg_bonuses_employee_id on vpg_bonuses (bonuses_employee_id);
```

```
create index idx_vpg_bonuses_payroll_id on vpg_bonuses (bonuses_payroll_id);
```

```
create index idx_vpg_bonuses_year_month on vpg_bonuses (bonuses_year,  
bonuses_month);
```

```
create index idx_vpg_bonuses_granted_at on vpg_bonuses (bonuses_granted_at);
```

-- Vacations indexes

```
create index idx_vpg_vacations_employee_id on vpg_vacations (vacations_employee_id);
```

```
create index idx_vpg_vacations_start_date on vpg_vacations (vacations_start_date);
```

```
create index idx_vpg_vacations_end_date on vpg_vacations (vacations_end_date);
```

```
create index idx_vpg_vacations_status on vpg_vacations (vacations_status);
```

-- Clock logs indexes

```
create index idx_vpg_clock_logs_employee_id on vpg_clock_logs  
(clock_logs_employee_id);
```

```
create index idx_vpg_clock_logs_timestamp on vpg_clock_logs (clock_logs_timestamp);
```

```
create index idx_vpg_clock_logs_log_type on vpg_clock_logs (clock_logs_log_type);
```

```
-- Labor events indexes
```

```
create index idx_vpg_employee_labor_event_employee_id on vpg_employee_labor_event  
(employee_labor_event_employee_id);
```

```
create index idx_vpg_employee_labor_event_labor_event_id on  
vpg_employee_labor_event (employee_labor_event_labor_event_id);
```

```
create index idx_vpg_employee_labor_event_start_date on vpg_employee_labor_event  
(employee_labor_event_start_date);
```

```
create index idx_vpg_employee_labor_event_status on vpg_employee_labor_event  
(employee_labor_event_status);
```

```
-- Users indexes
```

```
create index idx_vpg_users_username on vpg_users (user_username);
```

```
create index idx_vpg_users_email on vpg_users (user_email);
```

```
create index idx_vpg_users_national_id on vpg_users (user_national_id);
```

```
create index idx_vpg_users_role on vpg_users (user_role);
```

```
-- Report indexes
```

```
create index idx_vpg_report_logs_generated_by on vpg_report_logs  
(report_logs_generated_by);
```

```
create index idx_vpg_report_logs_report_type on vpg_report_logs  
(report_logs_report_type);
```

```
create index idx_vpg_report_logs_generated_at on vpg_report_logs  
(report_logs_generated_at);
```

```
create index idx_vpg_report_logs_status on vpg_report_logs (report_logs_status);
```

```
create index idx_vpg_report_versions_report_log_id on vpg_report_versions  
(report_versions_report_log_id);
```

```
create index idx_vpg_report_versions_created_at on vpg_report_versions  
(report_versions_created_at);
```

```
-- Audit logs indexes
```

```
create index idx_vpg_audit_logs_user_id on vpg_audit_logs (audit_logs_user_id);
```

```
create index idx_vpg_audit_logs_action on vpg_audit_logs (audit_logs_action);
```

```
create index idx_vpg_audit_logs_entity on vpg_audit_logs (audit_logs_entity);
```

```
create index idx_vpg_audit_logs_timestamp on vpg_audit_logs (audit_logs_timestamp);
```

```
-- === END OF INDEXES === --
```

